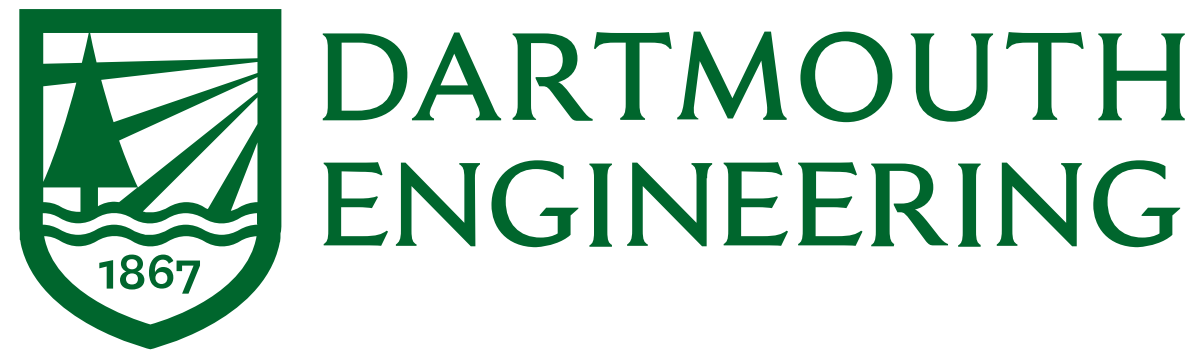


Joint-CS: Simultaneous Illumination Normalization and Sparse Image Compression via Unfolded Recovery

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Motivation

Cameras sample then throw away. A JPEG keeps only 5–10% of DCT coefficients, and the ISP further reshapes the data through demosaicing, tone-mapping, and white-balance. Compressed sensing reverses this: $M \ll N$ random measurements suffice if the signal is sparse [3, 4].

The sequential pipeline wastes its prior:

- RAW sensors measure $\boldsymbol{x} = \boldsymbol{s} \odot \boldsymbol{g}$, the *product* of sparse reflectance and a smooth illumination field.
- CS-then-correct enforces ℓ_1 on $\boldsymbol{x}$, but $\boldsymbol{x}$ is *not* DCT-sparse; multiplication by $\boldsymbol{g}$ destroys the sparsity the regularizer relies on.
- Post-hoc column-mean correction can rescale but cannot restore information already shrunk away.

Our Contributions

- Joint-CS objective.** A single ℓ_1 +smooth-prior problem in $(\boldsymbol{c}, \boldsymbol{g})$ solved by block-coordinate alternation: FISTA on $\boldsymbol{c}$, closed-form ridge on $\boldsymbol{g}$.
- Deep-unfolded inner solver.** A K =10-layer LISTA [6] with Adam, gradient clipping and best-checkpoint restore for stability.
- Scene-dependent benchmark.** Phase transition, multi-image rate-distortion, LISTA-vs-classical, and per-scene illumination ablation on 5 natural images.
- Honest failure-mode analysis.** Joint catastrophically fails on near-uniform scenes (moon, −8 dB); we trace it to rank-deficiency of $\boldsymbol{B}^\top \boldsymbol{B}$.

Big Takeaway

Estimating reflectance and illumination *jointly* beats the conventional CS-then-correct pipeline by up to +4.7 dB on textured scenes — *when* the scene has enough column-wise energy to identify the gain.

- Textured wins:** +1.1 to +4.7 dB (cameraman, astronaut).
- LISTA speedup:** K =10 matches FISTA at 22 iters.
- Failure mode:** −8 dB on near-uniform (moon) – needs an identifiability gate.

Background: Compressed Sensing

With $\boldsymbol{x}=\boldsymbol{\Psi}\boldsymbol{c}$ and $\|\boldsymbol{c}\|_0 \leq S$, the forward model is $\boldsymbol{y}=\boldsymbol{A}\boldsymbol{x}+\boldsymbol{w}$, $\boldsymbol{A} \in \mathbb{R}^{M \times N}$. If $\boldsymbol{A}$ satisfies the RIP with $\delta_{2S} < \sqrt{2}-1$, the Lasso relaxation

$$\boldsymbol{c}_\lambda = \arg \min_{\boldsymbol{c}} \frac{1}{2} \|\boldsymbol{A}\boldsymbol{\Psi}\boldsymbol{c} - \boldsymbol{y}\|_2^2 + \lambda \|\boldsymbol{c}\|_1$$

recovers $\boldsymbol{c}$ to within $O(\epsilon)$ [3]. We use the 2D DCT as $\boldsymbol{\Psi}$, a strong natural-image sparsifying basis.

The Sequential-Pipeline Problem

A conventional pipeline recovers $\boldsymbol{x}$ first, then divides out an illumination estimate. Two errors compound:

- Sparsity broken at the prior.** Pointwise multiplication by $\boldsymbol{g}$ kills DCT-domain sparsity, so ℓ_1 over-shrinks high-illumination coefficients.
- Correction is shape-rigid.** A column-mean divisor copies the gradient but cannot restore the information the over-shrinkage has already destroyed.

Joint-CS estimates $\boldsymbol{c}$ and $\boldsymbol{g}$ together so the ℓ_1 prior sees the true sparse signal $\boldsymbol{s}$.

Notation

$\boldsymbol{y} \in \mathbb{R}^M$	CS measurements ($M \ll N$)
$\boldsymbol{A} \in \mathbb{R}^{M \times N}$	Gaussian sensing matrix
$\boldsymbol{\Psi}$	2D DCT sparsifying basis
$\boldsymbol{c}$	sparse DCT coefficients
$\boldsymbol{s}=\boldsymbol{\Psi}\boldsymbol{c}$	sparse reflectance image
$\boldsymbol{g} \in \mathbb{R}^{\sqrt{N}}$	per-column illumination gain
$\boldsymbol{x}=\boldsymbol{s} \odot \boldsymbol{g}_{\text{pix}}$	RAW signal

Method 1: Joint-CS Objective

Replace the two-stage pipeline with a single objective:

$$\min_{\boldsymbol{c}, \boldsymbol{g}} \frac{1}{2} \|\boldsymbol{A} \text{diag}(\boldsymbol{g}_{\text{pix}}) \boldsymbol{\Psi} \boldsymbol{c} - \boldsymbol{y}\|_2^2 + \lambda_c \|\boldsymbol{c}\|_1 + \lambda_g \|\nabla \boldsymbol{g}\|_2^2$$

Jointly non-convex but **block-convex** in each variable.

c-step. Fix $\boldsymbol{g}$; effective sensing matrix $\boldsymbol{A}_{\text{eff}}=\boldsymbol{A} \text{diag}(\boldsymbol{g}_{\text{pix}}) \boldsymbol{\Psi}$, update is a standard Lasso solved by FISTA.

g-step. Fix $\boldsymbol{c}$; with $\hat{\boldsymbol{s}}=\boldsymbol{\Psi}\boldsymbol{c}$ the model is linear in $\boldsymbol{g}$, $\boldsymbol{y}=\boldsymbol{B}(\hat{\boldsymbol{s}})\boldsymbol{g}+\boldsymbol{w}$, and the update is closed-form ridge:

$$\boldsymbol{g}^* = (\boldsymbol{B}^\top \boldsymbol{B} + \lambda_g \boldsymbol{L})^{-1} \boldsymbol{B}^\top \boldsymbol{y},$$

with $\boldsymbol{L}=\nabla^\top \nabla$ the 1D discrete Laplacian. We clamp $\boldsymbol{g} \geq \epsilon_g$ to avoid divide-by-zero.

Gain ambiguity. The model has scalar invariance $(\boldsymbol{s}, \boldsymbol{g}) \rightarrow (a\boldsymbol{s}, \boldsymbol{g}/a)$; we resolve at evaluation by the optimal positive gain $a^*=\langle \hat{\boldsymbol{s}}, \boldsymbol{s}_{\text{true}} \rangle / \|\hat{\boldsymbol{s}}\|^2$, applied identically to baselines.

Method 2: Deep-Unfolded LISTA Inner Solver

Replace the FISTA $\boldsymbol{c}$ -step with K learned layers [6]:

$$\boldsymbol{c}^{(k+1)} = \mathcal{S}_{\theta_k}(\boldsymbol{W}_t \boldsymbol{c}^{(k)} + \boldsymbol{W}_e \boldsymbol{y}), \quad k = 0, \dots, K-1.$$

Initialise $\boldsymbol{W}_e=\frac{1}{L}(\boldsymbol{A}\boldsymbol{\Psi})^\top$, $\boldsymbol{W}_t=\boldsymbol{I}-\frac{1}{L}(\boldsymbol{A}\boldsymbol{\Psi})^\top \boldsymbol{A}\boldsymbol{\Psi}$, $\theta_k=\lambda/L$ to reproduce ISTA at step zero, then train end-to-end on $(\boldsymbol{y}, \boldsymbol{c})$ pairs.

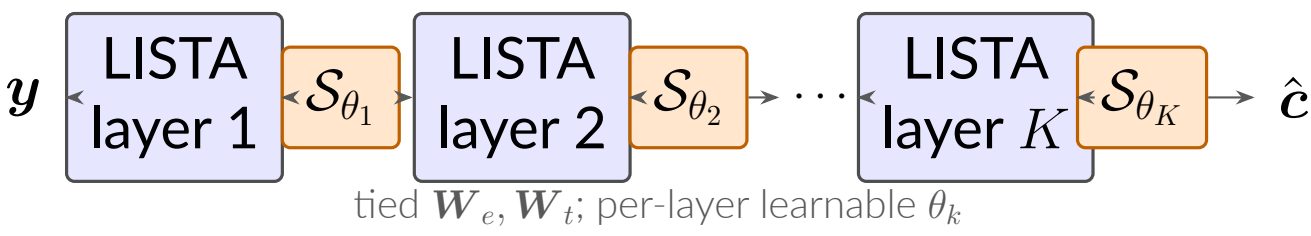


Figure 1. K -layer LISTA with tied weights and per-layer thresholds. Each θ_k adapts to that layer’s effective noise; the recurrence reproduces ISTA at initialisation.

Stability tricks

- Naive Adam@5×10^{−3}: clean descent to 0.12 NMSE, then blow-up to ∼10³.
- Diagnosis: $\rho(\boldsymbol{W}_t)>1$ along some eigendirection causes the recurrence to diverge.
- Fix: (i) lr→10^{−3}, (ii) clip-norm 1.0, (iii) restore best-val checkpoint.

Encoder → Decoder Pipeline

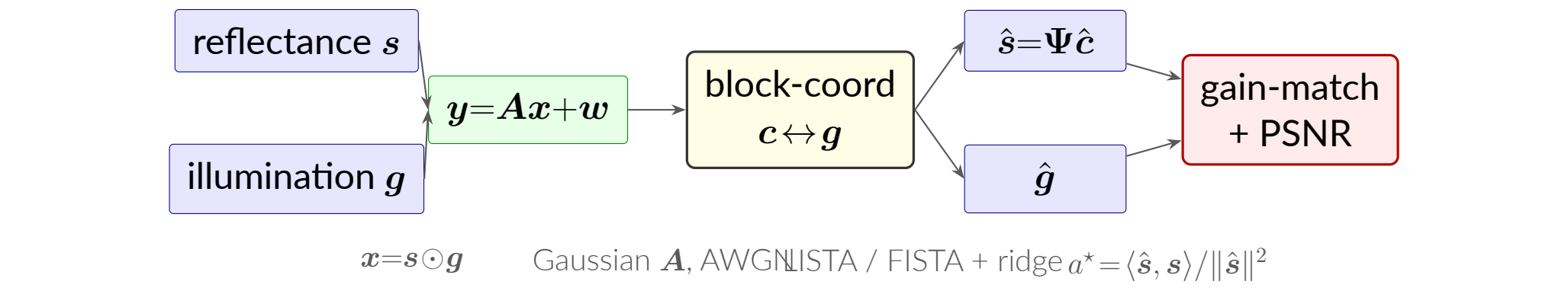


Figure 2. End-to-end Joint-CS. The RAW signal $\boldsymbol{x}=\boldsymbol{s} \odot \boldsymbol{g}$ is measured via Gaussian $\boldsymbol{A}$; block-coordinate alternation recovers $(\boldsymbol{c}, \boldsymbol{g})$ simultaneously; the scalar gain ambiguity is resolved at evaluation.

Block-Coordinate Recipe

- Initialise $\boldsymbol{g}^{(0)}=\mathbf{1}$, $\boldsymbol{c}^{(0)}=\mathbf{0}$.
- c-update.** Hold $\boldsymbol{g}^{(t)}$; run T_c FISTA (or K LISTA) steps on $\boldsymbol{A}_{\text{eff}}(\boldsymbol{g}^{(t)})$.
- g-update.** Hold $\boldsymbol{c}^{(t+1)}$; solve $(\boldsymbol{B}^\top \boldsymbol{B} + \lambda_g \boldsymbol{L})\boldsymbol{g}=\boldsymbol{B}^\top \boldsymbol{y}$ in closed form; clamp to $[\epsilon_g, \infty)$.
- Repeat $T_{\text{outer}}=8$ outer iterations.
- Resolve scalar gain ambiguity at evaluation.

Phase Transition (Sec. 3.2)

Empirical Donoho–Tanner boundary on synthetic Gaussian CS, $N=200$, 20 trials/cell, NMSE<10^{−3}.

$\delta = M/N$	0.30	0.50	0.70	0.90
OMC ρ^*	0.35	0.40	0.40	0.50
FISTA ρ^*	0.20	0.30	0.50	0.65

- OMP wins at low δ :** exact LS-on-support beats ℓ_1 when measurements are scarce.
- FISTA wins at high δ :** ℓ_1 keeps expanding the feasible region; OMP plateaus on dense supports.
- Crossover** near $\delta \approx 0.65$.

Rate-Distortion (Sec. 3.3)

5-image natural-scene set, SNR=30 dB, 16×16 blocks. Mean PSNR±std across scenes.

δ	0.20	0.40	0.50	0.70
OMP	17.4±4.3	20.4±3.7	21.7±3.2	24.9±3.0
FISTA	19.4±4.5	23.6±4.1	25.1±3.9	27.6±3.5
ADMM	17.6±4.8	23.5±4.4	25.0±4.0	27.5±3.7

Verdict. FISTA and ADMM agree within 0.5 dB above $\delta=0.2$ (same Lasso objective); both beat OMP by 2–3 dB. Wide std reflects scene difficulty: moon is easy (> 30 dB), coins is hard (∼20 dB).

LISTA vs. Classical at Matched Compute

- Setup.** $K=10$ tied-weight LISTA, 5000 synthetic $(\boldsymbol{y}, \boldsymbol{c})$ pairs, $N=200$, $M=80$, $S=10$, SNR=30 dB.

Solver	Iters/Layers	Val NMSE
ISTA	10	0.467
FISTA	10	0.337
LISTA	10	0.050
FISTA (matched)	22	≈ 0.05
FISTA (conv.)	500	0.005

2.2× **inference speedup** at matched recovery; LISTA still trails the converged ℓ_1 minimum by 10×, but amortises that gap into a 10× smaller cost at deployment.

Joint vs. Sequential: Per-Scene PSNR Gain (dB)

8× horizontal illumination gradient, SNR=25 dB, 16×16 center patch per scene. $\Delta=\text{PSNR}_{\text{joint}}-\text{PSNR}_{\text{seq}}$.

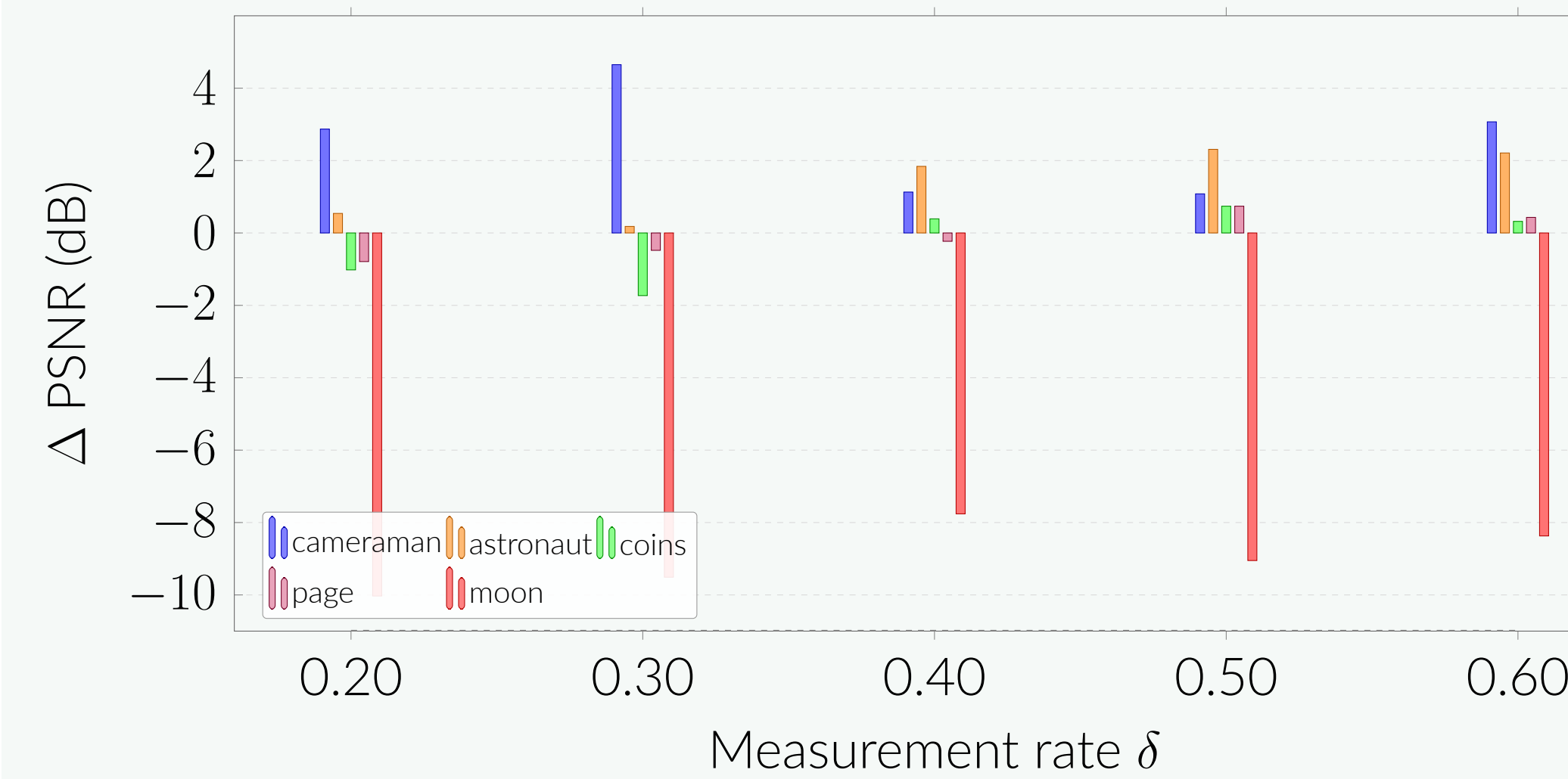


Figure 3. Per-scene PSNR gain of Joint-CS over the sequential baseline. Textured scenes (cameraman, astronaut) gain +1.1 to +4.7 dB; coins/page are roughly a wash; **moon collapses** by ∼8 dB – a real, mechanism-explainable failure mode.

Why Moon Fails (and Why That's Informative)

Moon’s centre patch is mostly black sky with a thin sliver of bright surface. After 8× gradient multiplication, most columns carry near-zero energy, so $\boldsymbol{B}^\top \boldsymbol{B}$ is severely rank-deficient and per-column gains are non-identifiable. The alternation then drives $\boldsymbol{g}$ to a high-variance solution that fits noise. The sequential pipeline avoids this by *never* estimating $\boldsymbol{g}$ from data – its column-mean divisor is essentially a copy of the gradient. **When the scene cannot identify $\boldsymbol{g}$, refusing to estimate it is a virtue.**

Deployment rule: gate the joint solution on $\kappa(\boldsymbol{B}^\top \boldsymbol{B})<\tau$; fall back to sequential otherwise. Not reported in prior CS+ISP literature.

Limitations & Future Work

- Scene-dependent failure** on near-uniform images (moon −8 dB) needs an identifiability gate.
- 1D illumination model** suits horizontal gradients; 2D vignettes need a richer $\boldsymbol{g}$ prior.
- Synthetic-trained LISTA** would need distribution-wide training for noise generalisation.
- Simulated measurements only;** physical coded-aperture validation remains future work.
- Outer-loop unfolding** (not just $\boldsymbol{c}$ -step) would yield spatially-adaptive thresholds.

References

[1] Amir Beck and Marc Teboulle. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM Journal on Imaging Sciences*, 2(1):183–202, 2009.
[2] Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. Distributed optimization and statistical learning via